

FINAL TERM EXAM ODD SEMESTER ACADEMIC YEAR 2025/2026
STUDY PROGRAM INFORMATICS
FACULTY ENGINEERING AND INFORMATICS
UNIVERSITAS MULTIMEDIA NUSANTARA

Subject	: IF351 – Database System	Date	:
Lecturer(s)	: Vincentius Kurniawan, Yustinus Widya, Budi Sitorus, Nofriyadi Nurdam, Florensia	Time	: 120 minutes
Form	: Open Book	Type	: Take Home

EXAM CONDITIONS / INSTRUCTIONS:

1. Read and understand the following questions carefully.
2. When answering, you are **allowed** to search for external sources except what is provided in e-learning and **allowed** to discuss or receive help from others.
3. File submission must be done through **UMN E-Learning** (according to the specified deadline)
4. **SQL** file naming format for each solution on each question must follow:
NIM_FullName_PartA_Solution.sql
Example: 00000012345_AmandaCitraDewanti_PartA_solution.sql
5. **JS / JSON** file naming format for each solution on each question must follow:
NIM_FullName_PartB_Solution.js/json
Example: 00000012345_AmandaCitraDewanti_PartB_solution.js/json
6. **PDF (for explanation)** file naming format for each solution on each question must follow:
NIM_FullName_Solution.pdf
Example: 00000012345_AmandaCitraDewanti_solution.pdf
7. ZIP those 3 (**SQL, JS, PDF**) files and file naming format for submission must follow:
NIM_FullName_ClassCode_UAS.zip
Example: 00000012345_AmandaCitraDewanti_IF351_AL_UTS.pdf
8. Double-check that your exam file has been submitted and that the uploaded file is correct.
9. **Plagiarism** and any form of cheating will result in a score of 0.
10. **Collaboration** is **strictly prohibited**. If answers are found to be identical, a score of 0 will be given.

DELIVERABLES:

1. A .sql file and .js or .json file containing all your queries/commands.
2. A short written explanation (**max 1 page**) describing your **design decisions**, especially:
 - How TCL commands ensure data consistency.
 - Why you used embedding or referencing.
 - What advantage your CTE or aggregation gives in performance or readability.

COURSE SUB-LEARNING OUTCOMES (SUB-CLO):

COURSE SUB-LEARNING OUTCOMES (SUB-CLO)		CLO
Codes	Descriptions	
Sub-CLO0712	Students are able to apply Transaction Control Language (TCL), manage users and privileges, create views, stored routines, triggers, and implement database security, including advanced SQL functions such as Common Table Expressions (CTE). (C3, C4)	CLO071
Sub-CLO0721	Students are able to perform CRUD operations (Create, Update, Delete) to manage NoSQL databases and use advanced NoSQL queries to optimize data processing.) (C3, C4)	CLO072

PROBLEMS Sub-CLO0712, Sub-CLO0721:

You are assigned as a **database developer** for a logistics company that manages shipments, drivers, and deliveries. The company uses both a **MySQL (relational)** database for financial transactions and a **MongoDB (NoSQL)** database for delivery tracking.

You must complete the following **integrated task**:

Part A — SQL Section (Relational Database)

- Create two related tables:
 - drivers (driver_id, name, total_saving, total_loan)
 - transactions (trans_id, driver_id, amount, type, trans_date)
- Implement a **TCL-controlled transaction** that updates a driver's total_saving and total_loan when a new transaction is inserted.
 - Use COMMIT, ROLLBACK, and SAVEPOINT appropriately.
 - Demonstrate at least **one rollback scenario** that ensures data consistency.
- Create a **trigger** that automatically updates the drivers table whenever a new record is inserted into transactions.
- Create a **view** called driver_balance_view showing each driver's name, total_saving, total_loan, and net balance.
- Create a **stored procedure** called get_driver_report(year) that returns the total transactions per driver for a specific year, using a **CTE** in your SQL.
- Grant privileges so that a user named finance_user can **SELECT** from the view but cannot modify the data.

Part B — NoSQL Section (MongoDB or similar)

1. Create a collection called deliveries with sample documents containing:

```
{
  "delivery_id": 101,
  "driver_id": 5,
  "route": "Jakarta-Bandung",
  "status": "completed",
  "packages": [
    {"package_id": "A12", "weight": 4.5, "status": "delivered"},
    {"package_id": "A13", "weight": 2.0, "status": "delivered"}
  ]
}
```

2. Perform CRUD operations to:
 - Insert at least **three** delivery records.
 - Update the status of one package to "returned".
 - Delete a delivery record with "status": "cancelled".
3. Using an **aggregation pipeline**, compute the **total delivered weight** per driver.
4. Explain (in your own words) **why you chose embedding or referencing** for the packages collection and how it affects query performance.

ASSESSMENT RUBRIC (per question):

Task	Criteria	Description	Points
Part A No. 1, 2	TCL Implementation	Correct use of COMMIT, ROLLBACK, and SAVEPOINT; demonstrates consistency control.	15
Part A No. 3, 4	Trigger + View	Trigger correctly updates totals; view correctly aggregates driver balances.	15
Part A No. 5	Stored Procedure (CTE)	Functional CTE used in the stored procedure; returns correct yearly totals.	15
Part A No. 6	Privilege Management	Correct GRANT/REVOKE commands implemented.	10
Part B No. 1	NoSQL CRUD	Correct insert, update, delete operations executed.	10
Part B No. 2	NoSQL Aggregation	Aggregation pipeline runs correctly and produces expected totals.	15
Part B No. 4	Data Modeling Explanation	Clear reasoning for embedding or referencing shows understanding of performance impact.	10
All Part A and Part B	Consistency & Documentation	Organized submission, clear code, readable comments, and concise explanations.	10
Total			100

This document has been approved by:

Arya Wicaksana (023971)
on 17/11/2025 12:22

